

AWS ETL Project: Social Media Data Processing using AWS Glue

1. Introduction

This project demonstrates the implementation of an ETL (Extract, Transform, Load) pipeline using cloud services. The goal is to process social media data (tweets and blogs), perform transformations such as joining datasets, extracting hashtags, and generating aggregated insights. The processed data is then stored for analysis.

2. Objective

- To build an ETL pipeline using AWS services
 - To extract data from Amazon S3
 - To transform data using AWS Glue Visual ETL
 - To load processed data into an S3 bucket
 - To analyze data using SQL queries
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3. Tools and Technologies Used

- Amazon S3 (Storage)
 - AWS Glue (ETL Processing)
 - AWS IAM (Access Management)
 - Amazon Athena (Querying)
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4. Architecture Overview

1. Data is uploaded into Amazon S3
2. AWS Glue Crawlers scan the data and create metadata tables
3. AWS Glue ETL job processes and transforms the data
4. Output is stored in another S3 bucket

5. Data is queried using SQL via Athena

5. Implementation Steps

5.1 S3 Bucket Creation

- Created bucket: etl-twitter-blog
 - Created folders:
 - etl-social-media
 - blog-data
 - Uploaded datasets:
 - sample_tweets.csv
 - sample_blogs.csv
 - Created output bucket: etl-cep-output
-

5.2 AWS Glue Database

- Database name: social_media_data
-

5.3 Classifiers

- twitter_data (CSV)
 - blog_data (CSV)
-

5.4 Crawlers

- tweet-crawl → tweets data
- blog-crawler → blog data

Tables created:

- sample_tweets_csv

- sample_blogs_csv
-

5.5 IAM Role

- Role: glue-role
 - Policy: AdministratorAccess
-

5.6 ETL Job (Visual ETL)

Transformations:

1. Join → user_id
 2. Drop Fields → remove duplicate column
 3. Regex Extractor → extract hashtags using #(\w+)
 4. Aggregate:
 - Count of tweets
 - Minimum timestamp
-

5.7 Output

- Stored in S3 bucket: etl-cep-output
-

5.8 Query Execution

Used Athena to run SQL queries:

```
SELECT user_id, COUNT(tweet_id)
```

```
FROM sample_tweets_csv
```

```
GROUP BY user_id;
```

6. Screenshots

6.1 S3 Bucket Creation

Buckets

General purpose buckets

All AWS Regions

Directory buckets

General purpose buckets (3) Info

Copy ARN

Empty

Delete

Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

< 1 >

	Name	AWS Region	Creation date
☐	aws-glue-assets-629017097546-us-east-1	US East (N. Virginia) us-east-1	April 24, 2026, 17:10:49 (UTC+05:30)
☐	cepoutput123	US East (N. Virginia) us-east-1	April 24, 2026, 14:55:14 (UTC+05:30)
☐	twitterblog	US East (N. Virginia) us-east-1	April 24, 2026, 14:47:39 (UTC+05:30)

Account snapshot Info

View dashboard

External access summary Info

6.2 Folder Creation and File Upload

cepoutput123 Info

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems - new

Access Points

Objects (1)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >

	Name	Type	Last modified	Size	Storage class
☐	 sample_blogs.csv	csv	April 24, 2026, 15:11:29 (UTC+05:30)	143.0 B	Standard

twitterblog Info

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems - new

Access Points

Objects (2)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >

	Name	Type	Last modified	Size	Storage class
☐	 blogdata/	Folder	-	-	-
☐	 socialmeadia/	Folder	-	-	-

6.3 Glue Database Creation

Databases (1)

Last updated (UTC)
April 24, 2026 at 12:05:39

Edit

Delete

Add database

A database is a set of associated table definitions, organized into a logical group.

Choose catalog

629017097546

Q Filter databases

< 1 > ⚙

☐	Name	Description	Location URI	Source catalog	Created on (UTC)
☐	socialmediadata	-	-	629017097546	April 24, 2026 at 09:27:40

6.4 Classifiers Setup

Classifiers (2) Info

Last updated (UTC)
April 24, 2026 at 12:06:43

Edit

Delete

Add classifier

View and manage all available classifiers.

Q Filter classifiers

< 1 > ⚙

☐	Name	Type	Classification	Last updated (UTC)
☐	blogdata	CSV	-	April 24, 2026 at 09:30:29
☐	twitterdata	CSV	-	April 24, 2026 at 09:29:14

6.5 Crawlers Configuration and Run

Crawlers (2) Info

Last updated (UTC)
April 24, 2026 at 12:10:09

Action

Run

Create crawler

View and manage all available crawlers.

Q Filter crawlers

< 1 > ⚙

☐	Name	State	Schedule	Last run	Last run ti...	Log	Table change...
☐	blogcrawler	✓ Ready		✓ Succeeded	April 24, 2026...	View log	1 created
☐	tweetcrawl	✓ Ready		✓ Succeeded	April 24, 2026...	View log	2 created

6.6 Data Catalog Tables

project jobs Last modified on 4/24/2026, 5:10:47 PM Actions Save Run

< **Visual** Script Job details Runs Data quality Schedules >

Data source - Data Catalog
blog

Data preview Info STARTING (2%) End session Previewing 0 of 0 fields

Name
blog

Database
Choose a database.
socialmediadata Refresh

Table
sample_blogs_csv Refresh

Use runtime parameters

Unsaved job found
We found an unsaved job, do you wish to Restore

Join Configuration

project jobs Last modified on 4/24/2026, 5:10:47 PM Actions Save Run

< **Visual** Script Job details Runs Data quality Schedules >

Transform - Join
Join

Transform - DropFields
Drop Fields

Custom prefix
Add a prefix to the field names of the parent node on the right
right Resolve it

Join type
Select the type of join to perform.
☒ Inner join
Select all rows from both datasets that meet ...

Join conditions
Select a field from each parent node for the join condition.

blog **tweet**
user id = user id Remove

Unsaved job found
We found an unsaved job, do you wish to restore it? Restore

Drop filled Configuration

project jobs Last modified on 4/24/2026, 5:10:47 PM Actions Save Run

Visual Script Job details Runs Data quality Schedules

Transform - DropFields
Drop Fields

Data preview Output schema

Data preview failure.
AccessDeniedException - User: arn:aws:iam::629017097546:user/odl_user_2196347 is not authorized to perform: glue:RunDataPreviewStatement because no identity-based policy allows the glue:RunDataPreviewStatement action

Node parents
Choose which nodes will provide inputs for this one.
Choose one or more parent node
Join
Join - Transform

DropFields

Field	Data type
☐ blog id	long
☒ user id	long
☐ timestamp	string

6.9 Regex Extractor Configuration

project jobs Last modified on 4/24/2026, 5:10:47 PM Actions Save Run

Visual Script Job details Runs Data quality Schedules

Transform - Dynamic Transform
Regex Extractor

Data preview Output schema

Data preview Info STARTING (2%)
End session Previewing 0 of 0 fields
Filter sample dataset

Transform

Name
Regex Extractor

Node parents
Choose which nodes will provide inputs for this one.
Choose one or more parent node
Drop Fields
DropFields - Transform

Column to extract from
String column on which to apply the regex.
tweet text

Unsaved job found
We found an unsaved job, do you wish to Restore

6.10 Aggregate Configuration

project jobs Last modified on 4/24/2026, 5:10:47 PM Actions Save Run

Visual Script Job details Runs Data quality Schedules

Transform - Aggregate
Aggregate

Data preview Output schema

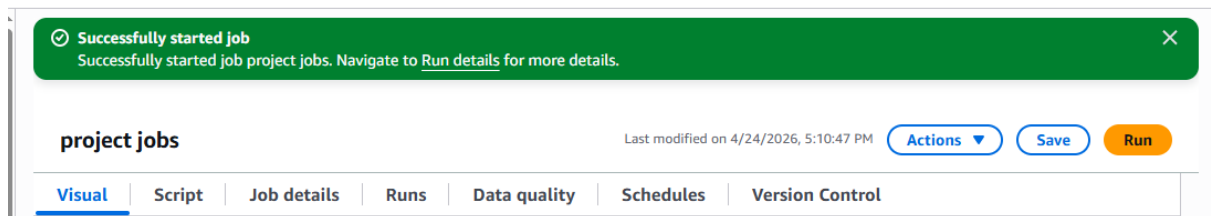
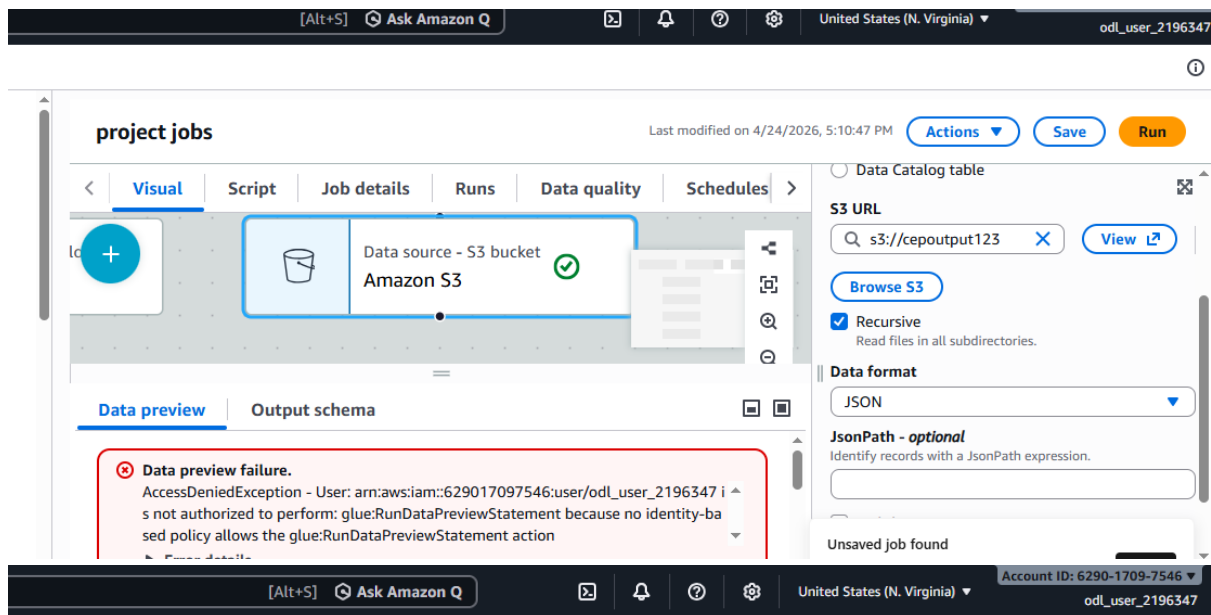
Data preview failure.
AccessDeniedException - User: arn:aws:iam::629017097546:user/odl_user_2196347 is not authorized to perform: glue:RunDataPreviewStatement because no identity-based policy allows the glue:RunDataPreviewStatement action

Field to aggregate
tweet id

Aggregation function Info
count

Field to aggregate
timestamp

Aggregation function Info
min



7. Results

- Successfully processed and joined datasets
- Extracted hashtags from tweets
- Generated aggregated insights
- Stored results in S3

8. Challenges Faced

- Issues connecting nodes in Visual ETL
- S3 permission errors
- UI glitches in Glue

9. Conclusion

The project successfully demonstrates building a cloud-based ETL pipeline using AWS services. It highlights efficient data processing, transformation, and storage.

10. Future Enhancements

- Real-time data streaming
 - Dashboard visualization
 - Automation using scheduling
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